

Literature Answer: Hyperbolic Groups in Candido–Cuth–Doucha

Question 1

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June 14, 2026

Status

Packet type: literature answer already present in later literature.

Verdict: Gartland’s arXiv:2406.10986 gives a positive answer, for every infinite finitely generated hyperbolic group, to Question 1 of Candido–Cuth–Doucha’s arXiv:1809.09957. The supporting paper explicitly states that its equation (1.2) answers [1, Question 1].

Source Metadata

Original source. Leandro Candido, Marek Cuth, and Michal Doucha, *Isomorphisms between spaces of Lipschitz functions*, arXiv:1809.09957v2; published in J. Funct. Anal. 277 (2019), 2697–2727.

Supporting answer source. Chris Gartland, *Hyperbolic Metric Spaces and Stochastic Embeddings*, arXiv:2406.10986v2.

Open Question

Candido–Cuth–Doucha ask:

Is $\mathcal{F}(\Gamma) \simeq \ell_1$ (and $\text{Lip}_0(\Gamma) \simeq \ell_\infty$) for every finitely generated hyperbolic group Γ ?

countable free group F' we have $\mathcal{F}(F') \simeq \ell_1$ and $\text{Lip}_0(F') \simeq \ell_\infty$ since F' isometrically embeds into an $\mathbb{R}$ -tree. This includes $F = \mathbb{Z}$ which is not even quasi-isometric to any non-abelian free group. So we face some restrictions. However, it leads us to pose the following question for hyperbolic groups which is motivated by the fact that asymptotic cones of finitely generated hyperbolic groups are (non-separable) $\mathbb{R}$ -trees (see e.g. [13, Proposition 11.167]).

Question 1. *Is $\mathcal{F}(\Gamma) \simeq \ell_1$ (and $\text{Lip}_0(\Gamma) \simeq \ell_\infty$) for every finitely generated hyperbolic group Γ ?*

Both positive and negative answer would be intriguing. Negative answer would show that the functor $\mathcal{F}(\cdot)$ can distinguish even some hyperbolic groups, while positive answer would imply that there exists a group Γ with property (T) with $\mathcal{F}(\Gamma) \simeq \ell_1$ - this follows since there are hyperbolic groups with property (T) (see e.g. [13, Chapter 10]). It is known that every action

Supporting Answer

Gartland's introduction states that the paper proves, for every infinite finitely generated hyperbolic group Γ ,

$$\mathrm{LF}(\Gamma) \approx \ell_1,$$

and immediately says that this answers [1, Question 1].

existing free space theory and the fact that $B_r(a)$ is uniformly (in $a \in A$) biLipschitz equivalent to a ball in $\mathbb{R}^n$ to obtain $\mathrm{LF}(\mathbb{H}^n) \approx \mathrm{LF}(\mathbb{R}^n)$.

In the final section, §7, we apply Theorem C and [12, Embedding Theorem 1.1] and prove in Lemma 7.1 that every infinite, finitely generated hyperbolic group Γ stochastic biLipschitzly embeds into a pointed $\mathbb{R}$ -tree and thus satisfies

$$(1.2) \quad \mathrm{LF}(\Gamma) \approx \ell^1.$$

This answers [16, Question 1]. In particular, we get Theorem E (via Theorem 7.2) as an immediate consequence. The unfamiliar reader may consult [43] for the definition of proper, uniformly Lipschitz affine actions.

Theorem E (Theorem 7.2). *For every infinite hyperbolic group Γ and finite generating*

The theorem-level statement appears again as Lemma 7.1.

action on a subspace of L^+ . As explained in §1.1, this line of research is inspired by a conjecture of Shalom ([17, Open Problem 14], [43, Conjecture 35]).

Lemma 7.1. *Every infinite, finitely generated hyperbolic group stochastic biLipschitzly embeds into a pointed $\mathbb{R}$ -tree and has free space isomorphic to ℓ^1 .*

Proof. Let Γ be an infinite, finitely generated hyperbolic group. Then Γ is uniformly discrete, and locally finite, and admits a rough biLipschitz embedding into $\mathbb{H}^n$ for some $2 \leq n \in \mathbb{N}$ by [12, Embedding Theorem 1.1]. Then the conclusion follows from Lemma 6.1. $\square$

Definitions

For a pointed metric space M , $\mathcal{F}(M)$ denotes the Lipschitz-free space over M , whose dual is canonically $\mathrm{Lip}_0(M)$, the Banach space of real-valued Lipschitz functions on M that vanish at the basepoint, equipped with the Lipschitz norm. Gartland writes $\mathrm{LF}(M)$ for the same free space. The notation $X \simeq Y$, or $X \approx Y$, means Banach-space isomorphism.

Proof Intuition

The original question asked whether the free space of a finitely generated hyperbolic group behaves, up to Banach-space isomorphism, like the free space of a tree. Gartland develops stochastic embeddings into $\mathbb{R}$ -trees and shows that such embeddings force the free space to be an L_1 -space; for infinite uniformly discrete hyperbolic groups, the resulting L_1 -space is ℓ_1 . Since $\mathrm{Lip}_0(\Gamma)$ is the dual of $\mathcal{F}(\Gamma)$, the corresponding Lipschitz-function space is then ℓ_∞ .

Formal Statement

Theorem. *Let Γ be an infinite finitely generated hyperbolic group, equipped with a word metric and any basepoint. Then*

$$\mathcal{F}(\Gamma) \simeq \ell_1 \quad \text{and} \quad \text{Lip}_0(\Gamma) \simeq \ell_\infty.$$

Proof. By [2, Lemma 7.1], every infinite finitely generated hyperbolic group stochastically bi-Lipschitzly embeds into a pointed $\mathbb{R}$ -tree and has Lipschitz-free space isomorphic to ℓ_1 . Gartland writes this as $\text{LF}(\Gamma) \approx \ell_1$; this is the same Lipschitz-free space denoted $\mathcal{F}(\Gamma)$ in [1]. Hence $\mathcal{F}(\Gamma) \simeq \ell_1$.

Taking Banach-space duals of an isomorphism $\mathcal{F}(\Gamma) \simeq \ell_1$ gives

$$\text{Lip}_0(\Gamma) = \mathcal{F}(\Gamma)^* \simeq \ell_1^* = \ell_\infty.$$

Thus both clauses of Question 1 hold for every infinite finitely generated hyperbolic group. $\square$

Scope And Limitations

The original question is phrased as “every finitely generated hyperbolic group.” Gartland states the result for every infinite finitely generated hyperbolic group and explicitly writes that it answers [1, Question 1]. This packet follows that intended scope. If finite groups are included literally, their free spaces are finite dimensional and therefore are not isomorphic to the usual infinite-dimensional ℓ_1 .

No new proof is claimed here; the packet records an already-known answer in a separate later paper.

Search And Verification Notes

Before packaging, the run indexes were searched for arXiv:1809.09957, the paper title, Lip_0 , Lipschitz-free spaces, and related keywords. No exact prior packet or attempt for this question was found; only broad rows for a different Lipschitz-free target appeared.

Bounded web search using exact phrases such as “Lipschitz-free space hyperbolic group ℓ_1 ” and “ $\text{F}(\Gamma)$ ℓ_1 hyperbolic group” located arXiv:2406.10986. The supporting paper cites Candido–Cuth–Doucha as reference [16], explicitly writes that equation (1.2) answers [16, Question 1], and gives the theorem-level statement in Lemma 7.1.

Human Review Recommendation

Verify that Gartland’s reference [16] is the Candido–Cuth–Doucha source and that the intended reading of Question 1 is the standard infinite-group reading used in Gartland’s statement. The mathematical identification itself is direct.

References

- [1] L. Candido, M. Cuth, and M. Doucha, *Isomorphisms between spaces of Lipschitz functions*, J. Funct. Anal. 277 (2019), 2697–2727; arXiv:1809.09957.
- [2] C. Gartland, *Hyperbolic Metric Spaces and Stochastic Embeddings*, arXiv:2406.10986.